Skills Summary

Compass-and-Straightedge Constructions: Plan, Justify, Check

Use this reference while completing your worksheet or when studying.

Skill 1 — Writing a construction as a plan

A plan is a numbered list of moves, and every arc step must answer three questions: where is the compass *point*, did the *opening* change, and what is the new point or line *called*?

The four plans everything else is built from: perpendicular bisector of a segment; bisector of an angle; copy of a segment; copy of an angle. Every later construction is one of these repeated or combined.

Legal tools only: unmarked straightedge and compass. No measuring.

Example: Plan the construction of the perpendicular bisector of a segment AB.
Step 1. The line must pass through two points that are each equidistant from A and from B, so the plan is: manufacture two such points with equal arcs, then join them.
Step 2. Step one: compass on A, opening more than half of AB, arc above and below. Step two: same opening, compass on B, arc above and below, crossing at M and N. Step three: straightedge through M and N. $\Rightarrow$ line $\boldsymbol{MN}$ is the perpendicular bisector

Try it: Write the plan for bisecting angle DEF. Name the two arcs that must share one compass opening.

check: arc from E cuts the rays at P, Q; equal arcs from P and Q meet at R; draw ray $\boldsymbol{ER}$

Skill 2 — Justifying a construction

A justification never points at the picture. It names what is equal *by construction* and then applies a theorem. Two moves cover almost everything:

- **Equal radii.** Every point on one arc is the same distance from its centre, because all radii of a circle are equal. This is where the equal sides come from.
- **SSS, then CPCTC.** Equal radii give three pairs of equal sides, so the triangles are congruent, and the angles or halves you actually wanted follow from corresponding parts.

Example: Given segment AB, arcs of radius AB centred at A and at B meet at C. Prove $\triangle ABC$ is equilateral.
Step 1. The two arcs were drawn with one unchanged opening, so the claim to prove is about radii — use the definition of a circle rather than the drawing.
Step 2. C is on the circle centred at A with radius AB, so $AC = AB$; C is on the circle centred at B with the same radius, so $BC = AB$. All three sides are equal. $\Rightarrow$ $\triangle ABC$ is equilateral

Try it: In the perpendicular-bisector construction the arc crossings are M and N. Name the congruent triangles and the criterion.

check: $\triangle AMN \cong \triangle BMN$ by SSS, then CPCTC

Skill 3 — Checking a construction on coordinates

Every construction promises one testable property. Name the property first, then test it with the distance formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Property by construction: midpoint or perpendicular bisector → two distances equal; copy of a segment → copy equals original; circumcentre → all three vertex distances equal; inscribed regular hexagon → side equals radius.

Example: A bisector construction on segment AB gives the crossing point $M(6, 4)$, and the endpoint A sits at $(2, 1)$. Test the construction at M.
Step 1. A perpendicular bisector promises equal distances to the two endpoints, so compute MA and then check the other endpoint the same way.
Step 2. $MA = \sqrt{(6-2)^2 + (4-1)^2} = 5$, and repeating the computation from the other endpoint returns the same radicand, so the two distances agree. ⇒ $MA = \mathbf{5}$, equal to MB

Try it: A bisector construction gives the crossing point $M(7, 5)$, and the endpoint A sits at $(1, 1)$. Compute MA to the nearest hundredth.

check: **7.21**